

10 For each box, since $\Delta V = 0$, there is no work done on the gases.

Thus $\Delta Q_{in} = \Delta U$ from the 1st law of thermodynamics.

Since the gases are ideal monatomic, $\Delta U = \frac{3}{2}nR\Delta T$.

Heat gained by argon = Heat lost by neon

$$\frac{3}{2}(1)R(T - T_o) = -\frac{3}{2}(2)R(T - 2T_o)$$

$$T = \frac{5}{3}T_o$$

11 Using ideal gas relation